

Hybrid Multi-Frequency Image Illusion

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Abstract—This project aims to create hybrid multi frequency image illusion by the use of classic image processing methods without any use of AI/ML models. This method follows the combination of two images, one is low frequency and the second is high frequency image in a weighted average which creates illusion of the image viewed at varying distance. The initial set of work primarily focuses on the gray-scale images and conducted the quantitative analysis for the same. The same method was also implemented for the color images, ensuring that the visual illusion is approximately close to the gray illusion image and more realistic. Additionally, we explored the demorphing for both grayscale and color images which ensures the reversibility of a frequency domain manipulation and gives insight about the effectiveness of classical image processing techniques for generating perceptual illusion images.

Index Terms—Hybrid image, frequency, Gaussian filter, image processing, visual illusion, quantitative analysis, low & high Frequency, illusion, morphing, colored image illusion, demorphing.

I. INTRODUCTION

Hybrid images illustrate a great visual phenomenon that is significantly leveraged on the visual system's sensitivity of the human eye to different spatial frequencies. The human eye tends to form a perception in which high frequency components of the image are able to dominate when viewed at close distances, while the human eye forms a perception where low frequencies are prominent when viewed from afar distances. This difference in perception depicts the foundation of the hybrid multi frequency image illusions. Unlike contemporary visual transformation techniques, which are based on learning models and generative approaches, these images can be constructed using traditional image-processing techniques. In this project, application of machine learning or artificial intelligence is intentionally avoided, focusing on mathematical and perceptual principles.

The project aims to design and analyse a complete hybrid image hybrid image pipeline with classical techniques only such as Gaussian filtering, frequency subtraction, morphing and weighted blending. Beginning with working on grayscale images by extracting low frequency components via smoothing and high frequency components through difference operations, the extraction were then combined with adjustable weights that created hybrid illusions which shifted on viewing depending on the scale. The morphing process ensured alignment within

the two images ultimately improving the clarity in the perception in the final hybrid image.

On development of a stable grayscale workflow, this technique was then applied to a dataset of 20-30 images for evaluation of its general applicability and stability across diverse image types. The project pipeline was further extended to developing hybrid illusion of colour images by independently performing frequency decomposition on each RGB channel. This helped in keeping the colour fidelity and retaining the hybrid illusion effect. Lastly, a de morphing or approximate reconstruction procedure was included for examining how much structural information remains within the hybrid image. Overall, this project demonstrates that without machine learning, the hybrid image illusions can be generated successfully. It gives a very powerful foundation for deep understanding of frequency-based visual processing and emphasises on further applicability of classical image-processing techniques in contemporary visual and perceptual practices.

II. METHODOLOGY

The method followed for creating hybrid multi frequency image illusion consist of three necessary steps, pre-processing, frequency, filtering, and combining image, further extended by the stage of demorphing to study the reversibility of the hybrid images. The method is as followed:

A. Preprocessing

In the first step, we give 2 input images of gray-scale or color and resize to same dimensions to make sure the spatial correspondence further, we normalize both the images to the range 021 and convert to floating point representation for accurate and precise frequency manipulation.

$$I = \frac{\text{image}}{255}$$

Here the alignment of both the images play a vital roll if they differ in the shape or structure. So, we apply precise affine morphing to ensure the facial or the objective features corresponds to each other. This is done so the low frequency image correctly overlap the high frequency details of image B.

B. Frequency Filtering

In second step, we apply question filter to both the images. First, we apply question filter with cut-off frequency σ_L to produce the low frequency component.

$$I_{low} = G_{\sigma_L}(I_A)$$

Similar by subtracting original to the blood version, we get high frequency component of image B

$$I_{high} = I_B - G_{\sigma_H}(I_B) \quad (1)$$

This technique helps to retain fine texture and edges essential for high frequency components.

C. Hybrid Image Combination

In this step, we combine the low frequency component of image A and the high frequency component of image B linearly, where k is a scale factor to adjust contrast of image B. Further output of this is clipped for valid intensity range.

$$H = G_{\sigma_L}(I_A) + k \times (I_B - G_{\sigma_H}(I_B)) \quad (2)$$

For color hybrids, equation (2) is executed independently on all three channels and then recombined into a single hybrid color image.

D. Demorphing / Reverse Frequency Extraction

As mentioned earlier to analyse the reversibility of hybrid images, the low frequency component of image A is reversed as,

$$H_{low} = G_{\sigma_L}(H)$$

while the high frequency component of image B is reversed as,

$$H_{high} = H - G_{\sigma_L}(H)$$

This method helps to get the insights that information of image A and B can be encoded in hybrid illusion. This experimental results of D morphing also validate the effectiveness of frequency, illusion and the extent to which the frequency band of hybrid image can be separated.

E. Color Image Illusion

Moreover, the same methodology is followed for the color image illusion as a grayscale image illusion. In the pre-processing step, we perform it separately on RGB channels while frequency filtering is also applied individually to all three channels. The step C and D remains same for the color images.

III. RESULTS

From the below experimental result, we obtain images Fig. 3 and 4 respectively, here figure 3 is gray-scale image illusion while figure 4 is color image illusion. Here we can observe that the gray-scale and color illusions are nearly same. Both the images show similar facial details when viewed at the near distance and show similar details of another facial structure at a far distance. From the below results, we can say that the intended perceptual illusion is highly effective. We see the low

frequency image at a far distance and high frequency image at near distance because the human eye perceptual diminishes the details at far distance.

A. Parameters to Change to Hybrid Image Quality Improvement:

Achieving good quality of a hybrid image is highly dependent on a number of tunable parameters associated with filtering, weighting, and preprocessing. These parameters are outlined in the following subsections and the effects they have on the end product. The sizes of Gaussian Kernels used in low-pass filtering are known as Gaussian kernel sizes (k).

1) *Kernel Size k of LPF1 (Low-Pass Filter applied to Image 1)*: The extent of smoothing of the first image is determined by the kernel size, which is $k = 65$. When you have a larger kernel, the LPF1 produced is less sharp which preserves much of the low frequency information and highlights the coarse structures in the resultant hybrid picture. On the other hand, a smaller kernel has more detail of Image1. (It is important to make sure that k is always an odd number.)

2) *kernel size k of LPF2 (to obtain HPF2 out of Image 2)*: The cutoff frequency of Image 2 to extract high-frequency content is defined by this kernel size (e.g., $k = 31$). Reading a smaller kernel only records very sharp edges, whereas a bigger kernel records slightly lower-frequency contents in addition to the high-frequency content. Based on blending, the weights of the two offer different selections of alpha (0.4, 0.6, 0.8) and beta (0.8, 1.0, 1.2). Weight α for LPF1: The degree of contribution made by the low-frequency representation of Image 1 is parameterised by the value of alpha. Higher values enhance the prevalence of low-frequency components that suggest Image 1 to be more prominent as one looks closer. This is minimised by smaller values. Weight β for HPF2: The parameter beta magnifies the input of the high-frequency fine-tuning of Image 2. The larger the values, the more the edges and sharp features will be visible (when viewed at a distance), whereas the smaller the value, the softer the high-frequency content.

Though it is usually 0 to 1, the values outside this range could also have interesting effects.

In the selection of input images, it is evident that the features of the object to be classified bear specific significance. Selection of input images. It is clear that the characteristics of the object, which will be classified, have a certain importance in the choice of input images. Though this is not coded as a parameter, the selection of input images is important. The structural information of the low-frequency image must be high whereas the high-frequency image must have the sharp details clearly. Images are better suited in that they should be of similar subjects and similarly oriented to perceive as hybrids.

The high degree of image alignment accuracy is due to the model's incorporation of both the local and global feature images. This is because the high image alignment accuracy is attributed to the fact that the model considers both the local and global feature images. Spatial alignment is

important in the development of hybrid image quality. Poor alignment because of poor landmark detection or transformation estimation may result in poor performance. Insurance of strong registration largely improves the clarity of hybrid images.

The experimentation procedure was carried out using the golf equipment. Note on Experimentation The experimentation process was performed with the aid of the golf equipment. An optimal hybrid effect needs to be accomplished through trial and error. Different choices of combinations of the size of kernels, weights of blending, and settings of alignment result in different perceptual behaviors at various viewing distances.



Fig. 1. Face 1



Fig. 2. Face 2



Fig. 3. Grey-scale hybrid image



Fig. 4. Colored hybrid image

IV. RESULT - DEMORPHING

The results of demorphing of illusion image shows the significance of the reversibility and the extent to which the original image can be recovered from the illusion image. Further, it also signifies the missing details and structure while

recovering the image, with potential sign of improvement on later stages.

V. DISCUSSIONS

VI. CONCLUSION

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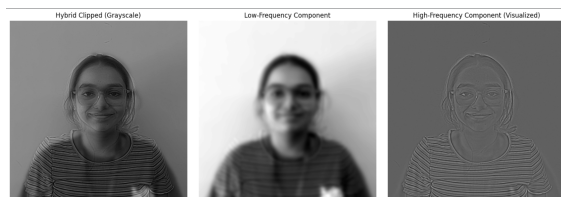


Fig. 5. Demorphed Illusion Image